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REMOTELY OPERABLE SURVEILLING AND SERVICE PROVIDING DEVICE, EMPLOYABLE ON MILITARY VEHICLES AND EQUIPMENT

Abstract

An angularly positionable sighting and surveilling device for a firearm mountable on military equipment including an angularly positionable firearm mount capable of being attached to a military equipment, a camera comprising a field of view attached to and in alignment with a barrel of the firearm, such that the camera is capable of providing sighting visuals for the firearm therefrom; and a transmitter capable of transmitting images captured by the camera to a remote display, and related devices and systems.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Patent Provisional Application No. 63/552,764, filed Feb. 13, 2024.

FIELD OF INVENTION

[0002] The present invention generally relates to the field of remotely operable military surveilling and service providing apparatuses. Employable on and in conjunction with, a broad range of military vehicles, and military equipment. employed by our armed forces. And capable of remotely providing military personnel, employing embodiments of them, with the ability to remotely maintain a full vertical, as well as a full horizontal visual, of the entire surroundings of the military vehicle, or military equipment. In which its employed, while additionally providing the military personnel employing embodiments thereof, with a plurality of additional needful surveilling and service providing abilities, from behind covered positions of safety, while doing so.

BACKGROUND

[0003] There have been numerous attempts to develop more efficient systems, and equipment, capable of providing our troops with a higher level of safety and control, in theaters of combat, that includes the risk of our troops being exposed to open fire by the enemy. Of which is a scenario that holds true, on a daily basis, in almost all military operations, needfully performed by our ground troops, in one way or another, to complete a mission required of them, and of which had historically, also resulted in a percentage of our troops never returning home to their family's and loved ones, when the conflict was over.

[0004] Unfortunately, in recent years. This scenario has proven to become even far more prevalent. This due to the many extremely dangerous scenarios confronting our troops, associated the many, all new dangers and challenges, needfully overcome by them, on a daily basis. In theaters of urban combat. Of which has proven to include many, extremely difficult challenges for our military leaders. That to this point, the prior art has fallen far short of effectively resolving. And unfortunately as a result, thereof, these problems remain responsible for a large percentage of U.S. Casualties. This simply due to the many advantages availed to the enemy, in theaters of combat of this nature. Of which the enemy combatants, are quick to take full advantage of, by taking up hidden and sniper positions, throughout the city. Resulting in our troops constantly having to expose themselves to the possible line of incoming enemy fire. Simply due to their inability to adequately acquire and maintain a visual of their entire surroundings and thus a knowledge of the true battlefield demographics and all of the hidden and otherwise, impeding dangers residing therein.

[0005] Of which, in recent years has presented, the greatest dangers and challenges, faced by our troops. This holding true in their performance of a broad range of duties, required of them, on a daily basis, In theaters of urban combat. Weather on watch, patrol, or reconnaissance mission of any kind, or simply when having to traverse from one position to another, or when having to clear enemy combatants and snipers from buildings that, they had been under fire from and having to view down passageways, and into various rooms and hiding places. Or in the performance intelligence gathering operations, and having to locate and properly assess enemy strongholds, troop movements and other developing enemy scenarios. All of which and more. constantly require our troops to expose themselves as a target, at various times and in various ways to effectively do so. This to mention only a few of a countless number of extremely dangerous battlefield scenarios and duties required of our troops. That have proven responsible for the largest percentage of U.S. casualties. However embodiments of the device. Included in this application, literally possess the ability to all but completely resolve. For any and all of our troops within military vehicles, or in the employment of military equipment or a type of military robotics, equipped with embodiments of

them. By providing them with the ability, to very safely and effectively perform, the greatest portion of the most dangerous duties required of them, from positions of full protective cover and without ever having to expose themselves as a target, at any time, or in any way, to very effectively do so. Of which the composition of all comprising embodiments of them, Capable of providing our troops with the ability to very effectively do so and more. Is included in and will be easily understood, by means of the detailed drawings and descriptions of the basic composition of all comprising embodiments of them. Included in this application

PRIOR ART

[0006] All prior art, found by the applicant, as a result of an exhaustive search performed by him, that was actually capable of performing a combination of even a few like service providing abilities, remotely comparable to those, providable by embodiments of the device. Included in this application. Not only acquired any of the actual sighting and, or surveilling services and abilities actually providable by them. In very different and far less effective ways. But any prior art, found by the applicant capable of providing a plurality of any like service providing abilities to those, providable by embodiments of the device. Included in this application. Could only do so and thus also actually included. Not only a plurality of apparatuses, capable of providing only individual, sighting, surveilling or other service providing abilities. Not only in the same or very similar ways, common to all prior art. But also in ways that also required and additionally included, a combination of complex additional electronic systems, equipment and components, to enable all to work together, as a whole. To possibly do so. Regardless of how, or where the composition of all function enabling components and apparatuses are housed or stationed on or within same.

[0007] Not only adding weight and production cost while also reducing the levels of dependability possible attainable by anything provided by the prior art. But also seriously limiting the actual range and types of military equipment and thus actual number of our troops in desperate need of any level of defensive safety and control providing abilities. Possibly providable by the prior art. From ever actually being provided to them.

[0008] However, neither do any of the embodiments of the device. Included in this application. Suffer from any of the limitations and inabilities, inherent to apparatuses and modes, employed by the prior art, to achieve and provide a plurality of the sighting and, or surveilling service providing abilities actually providable by them. But embodiments of the device. Included in this application. Also possess several desperately needed and extremely advantageous attributes and advanced surveilling and service providing abilities. Exclusive unto themselves Of which embodiments of the device. Included in this application achieve, by means of several unique and profoundly effective features arrangements and applicational modes, in which the composition of various individual and small combinations of the mechanical, electrical and electronic components. Comprising embodiments of them, are employed in, by and in conjunction with each other. That not only provides various portions of the embodiments of the device. Included in this application, with the ability to perform many of the sighting and surveilling abilities. Providable by embodiments of them, in ways and at levels, far superior to anything previously devised.

[0009] But that also provides a large portion of the composition of components. Comprising embodiments of them, with the ability to simultaneously, further employ each other's function enabling contributions, in ways. That not only expands the actual range of service providing abilities, providable by the composition of components, comprising embodiments of the device included in this application, much further. But that also provides embodiments of them, with the ability to perform the broad range of needful and extremely advantageous surveilling and service providing abilities. Providable by embodiments of them. By means of a radically reduced number of actual function enabling components, needfully employed in and by anything previously provided by the prior art capable of actually performing a plurality of any like service providing abilities. providable by embodiments of the device. Included in this application.

[0010] Additional prior art, referenced in FIG. 7-A. As employable in conjunction with

embodiments of the device. Included in this application, Isn't actually an apparatus of any kind at all. But rather a variation, of a type of software, included in a mobile application, developed approximately 10 years ago. Of which the applicant had always. Not only been aware of, but had also employed and thoroughly tested, over eight years ago, in conjunction with some of the first rough prototypes, constructed by the applicant, when proving concept for types of camera sights. Previously devised by him.

[0011] The information and instructions included within this mobile application, was initially developed either by, or for a company, that produced a device, that simply consisted of a smartphone mount, capable of securing a smartphone to the mounting rail of a paintball gun or firearm, called an Intelliscope. The mobile application developed by, or for the company, simply included instructions downloadable into the operating system of the smartphone, secured within the mount, capable of providing a superimposed reticle, as well as finger gesture controls, for the positioning of same in the screen of the smart phone and thus providing the user with the ability to employ the screen of the smartphone as a targeting screen. for the paintball gun or firearm in which its employed.

SUMMARY

[0012] Embodiments of the device. Included in this application. Include a type of remotely operable military surveilling and service providing apparatuses. Of which were devised and engineered, in all of the ways in which they were. To develop a device, capable of, effectively providing our troops, with several extremely advantageous attributes and advanced surveilling and service providing abilities, desperately needed by them. Of which embodiments of the device. Included in this application achieve, by means of several unique and profoundly effective features arrangements and applicational modes, in which the composition of various individual and small combinations of the mechanical, electrical and electronic components. Comprising embodiments of them, are employed in, by and in conjunction with each other. That not only provides various portions of the embodiments of the device. Included in this application, with the ability to perform many of the service providing abilities. Providable by embodiments of them, in ways and at levels, far superior to anything previously devised.

[0013] But that also provides a large portion of the composition of components. Comprising embodiments of them with the ability to simultaneously, further employ each other's function enabling contributions, in ways. That not only expands the actual range of service providing abilities, providable by the composition of components, comprising embodiments of them. Much further. But also provides embodiments of the device. Included in this application, with the ability to perform the broad range of needful and extremely advantageous surveilling and service providing abilities. Providable by embodiments of them. By means of a small percentage of the actual number of function enabling components, needfully employed in and by anything previously provided by the prior art, capable of actually performing a combination of even a few service providing abilities, remotely comparable to those providable by embodiments of the device. Included in this application. Not only do so in far less effective ways. But all have also only been capable of doing so, and thus actually comprise, of a combination of equipment, capable of performing only individual, or common, small task combinations. Of which, each additionally require and include a combination of complex additional electronic systems, equipment and components, to enable all to work together, as a whole. To possibly do so. Not only adding weight and production cost while also reducing the levels of dependability possible attainable by anything provided by the prior art. But also seriously limiting the actual range and types of military equipment and thus actual number of our troops in desperate need of any level of defensive safety and control providing abilities. Possibly providable by the prior art. From ever actually being provided to to them.

[0014] However, all aforementioned along with a full understanding of the many needful attributes and advanced service providing abilities, possessed by embodiments of the device. included in this

application, will be easily understood the following drawings and descriptions of embodiments of them.

[0015] In an embodiment, a device for mounting a firearm on a vehicle may include a mounting base, a housing portion, and a first and second bracket. The mounting base may include a base portion capable of being attached to the vehicle, and a rotating portion capable of rotating relative to the base portion. The housing portion may be disposed on the rotating portion of the mounting base. The first and second bracket may be capable of being secured to the firearm on opposite lateral sides of the firearm, and may further be secured to the housing portion.

[0016] In an embodiment, an angularly positionable sighting and surveilling device for a firearm mountable on military equipment may include an angularly positionable firearm mount capable of being attached to a military equipment, a camera comprising a field of view attached to and in alignment with a barrel of the firearm, such that the camera may be capable of providing sighting visuals for the firearm therefrom; and a transmitter capable of transmitting images captured by the camera to a remote display.

[0017] In an embodiment, a targeting and surveilling system for remotely controllable military equipment may include a remotely controllable firearm mount, a camera, and a set of controls. The remotely controllable firearm mount may be capable of being mounted on a military equipment, and further may be capable of providing angular adjustability of a firearm mounted within the remotely controllable firearm mount. The camera may include a field of view, and may be attached to the firearm and in alignment with the barrel of the firearm, such that the camera may be capable of providing both the sighting visuals for the firearm and the operating visuals for the military equipment. The set of controls may be capable of providing the user with the combined remote controllability both of the firearm mount and the military equipment on which the firearm mount is mounted.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] FIG. 1. Includes a drawing of a profile view, of an embodiment of the device, as it would appear as a complete unit, mountable on a type of military vehicle, equipment, or military robotics. (Not shown.) In this example of an embodiment thereof.

[0019] FIG. 2. Includes a drawing of an embodiment of two brackets, #44-B. and #44C. capable of providing a mechanical connection between the surveilling portions, of the device and the composition of components, housed within the housing portions of #25. Capable of providing vertical adjustability to embodiments of the device. Included in this application. In this example of an embodiment thereof.

[0020] FIG. 3. Simply includes a drawing of an embodiment of the device, as it would appear with the rotatably adjustable base portion thereof #27. Capable of providing horizontal adjustability to the embodiment of the device, mounted beneath an overhang, or under another portion of the military vehicle, equipment, or military robotics. In which this embodiment of the device is employed. In this example of an embodiment thereof.

[0021] FIG. 4. Simply includes a drawing of a fully self-contained embodiment of the device. Of which unlike the embodiments thereof. Included in FIG. 3 and FIG. 1. This embodiment thereof. Includes the greatest portion of an embodiment of the basic function enabling electrical and electronic composition of components, housed within the digital video image sensor housing portion. #1-C. Of this example of an embodiment of the device.

[0022] FIG. 5. Includes a drawing of an overhead view of an embodiment of a digital video image sensor housing portion, of an embodiment of the device. Included in this application. As if this portion of an embodiment of the device, was transparent to enable the easy viewing of an example

of an employable type and arrangement, of a large portion of the basic composition of function enabling components. Optionally employable within this portion of an embodiment of the device. In this example of an embodiment thereof.

[0023] FIG. 6. Includes a drawing of an overhead view, of another one of many optional embodiment variations, of the digital video image sensor housing portion 1-B. of an embodiment of the device. Of which is also drawn, as if the housing portion. Of this embodiment thereof was also transparent, to enable the easy viewing of the composition of component housed within the digital video image sensor housing portion of an embodiment the device. In this example of an embodiment thereof.

[0024] FIG. 7-A. Includes a drawing of an overhead view of the bottom portion of an additional embodiment of a digital video image sensor housing portion of an embodiment of the device. Included in this application. As well as the composition and arrangement of components housed therein. In this example of an embodiment thereof.

[0025] FIG. 7-B. Simply includes a drawing of an embodiment of a removable access plate. #67. Providable for and attachable to the bottom portions of embodiments of the digital video image sensor housing portions of embodiments of the device. Included in this application. Simply included, both to enable the easy assemblage. as well as the possibility of feature access to the composition of components housed within same. In this example of an embodiment thereof.

[0026] FIG. 7-C. Also Includes a drawing of an overhead view of the bottom portion of an additional embodiment of a digital video image sensor housing portion of an embodiment of the device. As well as the composition and arrangement of components housed within this embodiment thereof. Of which actually shares many things common with the embodiment thereof. Included in FIG. 7-A. with the exception of this embodiment thereof being equipped with a plurality of digital video image sensors. In this embodiment thereof.

[0027] FIG. 7-D. Includes a drawing of an overhead view of an embodiment of the bottom portion of a rear viewing digital video image sensor surveilling device. Of which, would be simultaneously employable in conjunction with and employ the same military digital display device and either be made as an integral aft portion of an embodiment of the device itself, or mounted to an aft portion of the military vehicle equipment, or military robotics. In which the embodiment of the device is employed. In this example of an embodiment thereof.

[0028] FIG. 8. Includes a drawing of an embodiment of a portion of the device. Capable of providing an embodiment thereof. Both with a full vertical angle of adjustability, as well as 360 degrees of horizontal angular adjustability. Relative to the military vehicle, equipment or military robotics. In which the embodiment of the device is employed. In this example of an embodiment thereof.

[0029] FIG. 9. Includes a drawing of a blown-up view, of the same embodiment of the portions of the remotely operable military surveilling and service providing apparatus, as is included in FIG. 8. Of which is drawn, as if the outer housing portions of #25. And #27. Were transparent, to allow the composition of electrical and mechanical components housed within this embodiment thereof, to be easily viewed. In this example of an embodiment thereof.

[0030] FIG. 10-A. Includes an overhead view of the same embodiment of a mounting base #44-A. Along with the same first housing portion of #27. As is included in FIG. 9. Along with an embodiment of the basic composition of components that would remain within this portion of #27. As if same, had been removed from an embodiment of the second housing portion of #27. By means of the removal of mechanical fasteners #28-B. Included in the drawing thereof in FIG. 10-B. In this example of an embodiment thereof.

[0031] FIG. 10-B. Includes an overhead view of the same embodiment of the second housing portion of #27. As is included in FIG. 9. Of which is an integral portion of #25. In this embodiment thereof. This drawing of an embodiment thereof, also includes the composition of components that would remain within this embodiment of same. Once removed from the first housing portion of

#27. Included in FIG. 10-A. By means of the removal of the mechanical fasteners #28-B. In this example of an embodiment thereof.

[0032] FIG. 11. Includes a drawing of a profile view of the same portion of an embodiment of the device, as is included in FIG. 8. Through FIG. 10-B. As if the first and second outer portions of 25. As well as the first and second outer portions of 27. Had been re-assembled. But were also transparent. To enable the composition of components comprising the inner portions of an embodiment of the device to be easily viewed. In this example of an embodiment thereof.

[0033] FIG. 12-A. Simply includes a drawing of an embodiment of one of two, very common types and variations of remotely employable controls and display devices #59. Of which either include wireless, direct wire and, or fiber optic digital signal transferring and digital video image receiving components. In this example of an embodiment thereof.

[0034] FIG. 12-B. Includes a drawing of an example of an embodiment, of a wireless portable military display device. And set of controls. Employable in conjunction with embodiments of the device. Included in this application. Capable of providing the soldier or squad, employing the embodiment thereof, with the portable and mobile provision of all of the needful services and abilities, providable by the embodiment of the device, from remote positions, of protective cover and totally out of the possible line of incoming fire. In this example of an embodiment thereof.

DETAILED DESCRIPTION

[0035] Embodiments of the device. Included in the following drawings and descriptions. were devised and engineered, in all of the ways in which they were. To develop a single device that not only possessed several extremely important attributes, and needful service providing abilities, exclusively providable by embodiments of them. But to also devise and engineer, embodiments of them, in ways, that would additionally provide embodiments thereof, with the ability to perform, all of the needful, safety, surveilling and additional service providing abilities. Including those exclusively possessed by embodiments of them. Not only at levels of versatility, adaptability and efficiency. Impossibly attainable by any apparatus or combination of apparatuses, previously devised. But, also with the ability perform all providable by embodiments of them, Not only by means a single system. But also, by means of a small percentage of the actual function enabling components, needfully employed by anything previously devised, to perform the few sighting and service providing abilities actually providable by the combination of apparatuses and systems, employed by them. In their effort to effectively do so.

[0036] The modes in which embodiments of the device. Included in the following descriptions and drawings acquire their unique abilities to do so. Is actually attributable to a combination of several unique modes in which various portions of the composition of mechanical, electrical and electronic components. Comprising embodiments, of them, were devised and engineered arranged and employed that enables all comprising embodiments of them, with the ability to further employ each other's function enabling contributions, in very unique ways, that provides various portions and combinations, of the basic composition of additional components, comprising embodiments of them, with the ability to work in conjunction with and relative to each other, in ways that additionally enables various portions thereof, with the ability to further employ each other's function enabling contributions. In very unique ways, that, not only provides individual components, comprising embodiments of them, with the ability to simultaneously perform a plurality of basic function enabling contributions to the overall functionality of the embodiment of the device, as a whole.

[0037] But in ways that additionally provides various portions of the composition of components comprising embodiments of them, with several needful and highly advantageous, service providing abilities and combinations thereof providable by embodiments of them. In ways and at levels. Impossibly achievable, by any apparatus previously devised. Or anything develop-able in the future, for that matter, without constructing certain portions, of them in the same manner while also employing the same basic function enabling composition of components. Comprising various

portions of embodiments of them, in the same basic unique manner, in which their employed within and by embodiments of the apparatus. Included in this application.

[0038] In fact not only are embodiments of the device. Included in this application capable of performing several service providing abilities at levels impossibly attainable by anything previously devised. But any remotely comparable equipment or apparatuses provided by the prior art, capable of performing some of the basic function enabling services and abilities. Providable by embodiments of the device. Included in this application. Actually not only consist of a plurality of apparatuses, capable of performing only common service providing abilities, that require and additionally include, a plurality of complex electronic systems and components, added to and employed in conjunction with them, to provide each, with the ability to work in conjunction with each other, to possibly do so.

[0039] But as a result thereof. Also rendering it, impossible, for the prior art, to ever achieve the ability to provide a level or range of remotely comparable service providing abilities. At anywhere near the levels of versatility, adaptability or range of effective employability. Providable by embodiments of the device. Included in this application. Of which will also enable the operationally effective employment of embodiments of them. Along with the provision of the broad range of needful and extremely advantageous service providing abilities, providable by embodiments of them. On a large number, types and variations of military vehicles, equipment and military robotics. Presently employed by our troops, As well as others presently being developed for them. That will not only expand the roles of employment, and radically advance the levels and range of needful defensive safety and service providing abilities, providable by the military vehicle, equipment, or military robotics. In which their employed.

[0040] But will also provide the soldier or squad employing embodiments of them. Not only with the ability to maintain a full visual of their entire surroundings and maintain positional awareness, along with an ongoing knowledge of the true battlefield demographics and of the otherwise impending and hidden dangers, residing therein, while additionally providing them with the broad range of desperately needed and highly advantageous service providing abilities. Providable by embodiments of them, from positions of total protective cover, without any of our troops having to expose themselves to incoming fire, at any time, or in any way, while employing embodiments of them. But as result thereof provide our troops with the highest possible level of safety and control providing abilities, in their performance of a countless number of tasks and duties, required of them, on a daily basis. That have previously proven responsible for a large percentage of U.S. casualties.

[0041] Of which, along with a full understanding of the modes in which embodiments of the device. Included in this application, achieve all of the advanced service providing abilities. Providable by embodiments of them will all be easily understood. By means of the detailed descriptions and drawings. Included in the following.

[0042] Accordingly, the components have been represented where appropriate by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present invention so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.

[0043] In an embodiment, a device for mounting a firearm on a vehicle may include a mounting base, a housing portion, and a first and second bracket. The mounting base may include a base portion capable of being attached to the vehicle, and a rotating portion capable of rotating relative to the base portion. The housing portion may be disposed on the rotating portion of the mounting base. The first and second bracket may be capable of being secured to the firearm on opposite lateral sides of the firearm, and may further be secured to the housing portion.

[0044] In certain embodiments, the mounting base further comprises a crown gear capable of being turned by a shaft to effectuate the rotation of the rotating portion. In certain embodiments, at least one of the first and second bracket may include a second rotating portion capable of rotating the

firearm the relative to housing portion. Certain embodiments, may further include a camera affixed to the firearm, wherein the camera is aligned to capture images in a field of view centered in the firearm's line of fire. In certain embodiments, the field of view provided by the camera affixed to the firearm may be capable of also providing operating visuals for the vehicle. Certain embodiments may further include a transmitter capable of transmitting images captured by the camera to a remote display.

[0045] Certain embodiments may further include a set of controls comprising a transmitter capable of transmitting signals to the device providing a user with remote control over the device and the vehicle.

[0046] In an embodiment, an angularly positionable sighting and surveilling device for a firearm mountable on military equipment may include an angularly positionable firearm mount capable of being attached to a military equipment, a camera comprising a field of view attached to and in alignment with a barrel of the firearm, such that the camera may be capable of providing sighting visuals for the firearm therefrom; and a transmitter capable of transmitting images captured by the camera to a remote display.

[0047] In certain embodiments, a field of view provided by the camera attached to and in alignment with the barrel of the firearm is additionally may be capable of providing operating visuals for the military equipment in which the device is employed. In certain embodiments, the angularly positionable sighting and surveilling device for a firearm may additionally include with a set of controls capable of providing the user with the remote control of the angularly positionable sighting and surveilling device and the military equipment.

[0048] Certain embodiments, may further include a mounting base portion. Certain embodiments may further include a first rotatable portion attached to the mounting base portion of the device and capable of providing horizontal angular adjustability to the angularly positionable firearm mount relative to the mounting base portion of the device. Certain embodiments may further include a second rotatable portion attached to the first rotatable portion. In certain embodiments, the second rotatable portion may be capable of providing vertical angular adjustability to the angularly positionable firearm device relative to the mounting base portion of the device and the military equipment in which its employed.

[0049] In an embodiment, a targeting and surveilling system for remotely controllable military equipment may include a remotely controllable firearm mount, a camera, and a set of controls. The remotely controllable firearm mount may be capable of being mounted on a military equipment, and further may be capable of providing angular adjustability of a firearm mounted within the remotely controllable firearm mount. The camera may include a field of view, and may be attached to the firearm and in alignment with the barrel of the firearm, such that the camera may be capable of providing both the sighting visuals for the firearm and the operating visuals for the military equipment. The set of controls may be capable of providing the user with the combined remote controllability both of the firearm mount and the military equipment on which the firearm mount is mounted.

[0050] Certain embodiments, may further include a transmitter capable of receiving and transmitting video images provided by the camera. Certain embodiments may further include a display capable of receiving and displaying video images provided by the transmitter. In certain embodiments, the remotely controllable firearm mount may additionally include a first rotatable portion capable of providing horizontal angular adjustability to the angularly positionable firearm mount. In certain embodiments, the firearm and the camera may be disposed upon a second rotatable portion of the firearm mount, capable of providing vertical angular adjustability to the firearm and camera.

[0051] FIG. 1. Includes a drawing of a profile view, of an embodiment of a remotely controllable device, as it would appear, mountable on a type of military vehicle, equipment, or military robotics. (Not shown.) Of which includes an embodiment of the digital video image sensor housing #1-A.

Portion of this embodiment of the device, as if this embodiment thereof, was transparent, to enable the easy viewing of the basic composition of a large portion of the components housed within this embodiment of same. Of which all embodiments of the actual digital video image sensor housing portions of the device. Included in this application. May house any portion of all, of the actual electrical and electronic, function enabling component portions, of an embodiment of the device, In addition to the digital video image sensor. #2-A. and any additional component portions of an embodiment of the device needfully housed within the actual digital video image sensor housing portion of the particular embodiment of the device.

[0052] The actual digital video image sensor. #2-A. Portion of all embodiments of the device. Included in this application. Like the composition of a large portion of the rest of the components, comprising embodiments of the device. Are employed within embodiments of them, in a very unique manner, that provides the single digital video image sensor portion, of all embodiments of the device, Not only with the ability, to simultaneously perform a plurality of service enabling tasks performable by an embodiment of the device itself. But also for the military vehicle, equipment, or military robotics. In which the embodiment of the device is employed, while doing so. This along with additional service providing abilities, providable by the digital video image sensor portion of the device. enabled by means of the composition of all of the additional components, comprising embodiments of the device. Of which additionally provides same with a full vertical, as well as a full horizontal field of view of 360 degrees. along with the variably magnify ability of the field of view of the digital video image sensor portion of same, to any desired level of magnification, along with the provision of all needful additional service providing abilities. Providable by the embodiment of the device, 24-7. By means of very effective night viewing and, or thermal optic viewing abilities. Also provided to both, by means of the same digital video image sensor portion of the embodiment of the device, mounted on the military vehicle, equipment, or military robotics, in which the embodiment of the device is employed.

[0053] This embodiment of 2-A. comprises of a type of digital video image sensor that possesses, both optical zooming as well as infrared viewing abilities. And of which is provided with lower tech, but effective night viewing abilities, simply by means of the employment of the infrared viewing abilities of 2-A. Employed in conjunction with #11. Of which comprises an embodiment of an infrared illuminator, included in this drawing of an embodiment thereof. Of which has proven to be very effective when employed in conjunction with a significantly powered, variable infrared illuminator and may prove to be an advantageously employed. In some military applications of the device. However, most embodiments thereof, would be equipped with much higher tech thermal optic and other night viewing components, presently employed by our armed forces. Of which have also proven to be very effectively employable within embodiments of the digital video image sensor housing or other portions, of an embodiment of the device. Included in this application.

[0054] An embodiment of a composition of electronic components. Capable of effectively providing the digital video image sensor portion of an embodiment of the device, with thermal optic viewing abilities, are included in the drawings of embodiments thereof, included in FIG. 3. Through FIG. 7. And actually, viewable in the drawings of the embodiments thereof included in FIG. 5. And FIG. 6. as well as FIGS. 7-A through 7-D. This particular embodiment thereof, is shown as employing a standard type of reticle #14. As is employed in the greatest portion of almost all regular reflex type sights. Provided by means of a standard led type reticle projector #19. In this embodiment thereof. Powered by a button battery. (Not viewable in this drawing). However, of which would be housed within this embodiment of a reticle selector #21. In this embodiment thereof. The actual reticle included in this drawing of an embodiment of the device, is produced by means of the reflected image of the tiny light beam #20. Off of a slightly reflective inner surface portion of this embodiment of the digital video image sensors viewing window #10-A. located in direct linear alignment with the center of the lens portion #5-A. Of the digital video image sensor. #2-A. In this drawing of an embodiment thereof. #15. Included in this drawing of an embodiment

of the device. Comprises an adjustment mechanism, affixed to a portion of the digital video image sensor housing portion. Of this embodiment of the device. Of which intern comprises a pivotable connection to the mounting portion #9. of the digital video image sensor housing portion of this embodiment of the device. #4-A. and #4-B. Comprise manually employable control knobs. Capable of adjusting the sighting angle, of #15. And thus to the digital video image sensor housing portion of the device, along with the digital video image sensor and reticle, housed therein. Relative to the mounting portion of same #9. And the mounting rail #12. Of this embodiment of the device. Of which, actually includes only one of many employable mechanisms, known of the applicant, capable of performing these tasks that may be employed in other embodiments of them. However, for easy reference and convince purposes, the same basic type of angular adjustment knobs and mechanisms, were also included in the drawings of many of the other embodiments of the device. Included in this application.

[0055] #44-B. comprises an embodiment of one of two plates. Comprising an integral shaft portion of same. (Not viewable in this drawing on an embodiment of the device). However, of which comprises a mechanical connection to a gear. Housed within one of the inner housing portions, of this embodiment of #25. Of which comprises a portion of the composition of the mechanical, and electrical components, housed within one of the two vertical housing portions of #25. Included in this embodiment of the device. Capable of providing the device, #17 with a full range of vertical adjustability. Relative to the military vehicle, equipment or military robotics, in which the embodiment of the device is employed. Of which is indicated by means of the curved arrow #35. Included in this drawing of an embodiment thereof.

[0056] Both portions of this embodiment of #25. Are integral Portions of #27. Of which is an embodiment of a rotatably adjustable base portion of this embodiment of the device. Capable of housing the composition of mechanical and electrical component portions, of this embodiment of the device, capable of providing the embodiment of the device, with a full horizontal range of angular adjustability. Relative to the military vehicle, equipment, or military robotics. In which this embodiment of the device is employed. As indicated by means of the curved arrow #34. Included in this drawing, of an embodiment thereof. The basic composition of components housed within #25 and #27. Portions of this embodiment of the device, are included in the drawings and descriptions of this embodiment thereof. Included In FIG. 9. through FIG. 11.

[0057] This drawing of an embodiment of the device also includes digital video image transmitter. #3-A. portion of this embodiment of the device. Housed within the digital video image sensor housing portion of this embodiment of the device. Of which, may, or may not actually be housed within this portion, of other embodiments of the device. Of which, would also hold true for many of the additional basic function enabling electrical and electronic components, comprising embodiments of the device. Not necessarily housed within the actual digital video image sensor housing portion of the embodiment of the device, to enable the overall functionality and employability of embodiments of the device, as a whole, on the actual type of military, equipment, vehicle or military robotics. In which embodiments of the device. Included in this application, would be employed.

[0058] In that even though, the basic composition of the greatest portion of the function enabling components. Comprising embodiments of the device. Would be provided for all embodiments of them. The composition of which may or may not be housed within the actual digital video image sensor housing portion of the particular embodiment of the device itself. In that, weather due to a practical advantage attainable. Relative to the actual type of military equipment, in which the embodiment of the device is employed, or relative to other equipment or systems employed on and by same. like the embodiment thereof, included in this drawing. As well as several others. Included in this application. Portions of the basic function enabling composition of components, comprising embodiments of the device. May be stationed on, or within another portion or portions, of an embodiment of the device. Configured in a manner to accommodate same, or other. To achieve a

higher level of operational efficiency, or to most effectively provide all of the needful services and abilities providable by the embodiment of the device, on the particular type of military vehicle, equipment or military robotics. In which the embodiment of the device is employed.

[0059] The composition of all electrical and electronic component portions, of this embodiment of the device. Including all components viewable in this drawing of an embodiment thereof, as well as the composition of all, not actually viewable in this drawing of an embodiment thereof. Would maintain all needful electrical and digital communication and enable the interactional relationship between the composition of all electrical and electronic function enabling components. Comprising this, as well as several other embodiments of the device. Included in this application. By means of, direct wire and, or fiber optic communication and of which would be transferred between the various portions thereof, by means of a plurality of wires and, or tiny fiber optic cables within cable #8. included in this drawing of an embodiment of the device.

[0060] Some of the additional, separately stationed, function enabling electrical and electronic components that may be included in this embodiment of the device. But are not actually viewable in this drawing of an embodiment thereof. Are included in some of the other drawings and descriptions of other embodiments of the device and portions thereof. Included in this application. Such as the composition of the additional function enabling electrical and electronic components. Included in the drawing and description of an embodiment of a digital video image sensor, housing portion of an embodiment of the device and the composition of the electrical and electronic, components housed within same. Included in FIG. 5. Of which also Includes the actual power supply for the embodiment of the device. As well as additional component portions thereof, actually housed on, or within the digital video image sensor housing portion of the embodiment of the device. Such as an omnidirectional transmitting antenna. Capable of transmitting all digital video images,

[0061] received from the transmitter #3-B. Portion of the embodiment of the device, to a receiving antenna and receiver portion of a remotely employable digital display or digital display device, employed in conjunction with the embodiment of the device. (Also Not included in this drawing.) However, a drawing of a couple of embodiments thereof, is Included in FIGS. 12-A and FIG. 12-B. Of which also includes sets of controls. Of which would also include a transmitter and transmitting antenna Capable of transmitting digital signals from same, to an omnidirectional receiving antenna and receiver portion of the embodiment of the device, (also not included in this drawing of an embodiment thereof.) Either, by means of direct wire, fiber-optic, or by means of wireless digital communication. As well as additional function enabling components. Such as switching and power distribution components. Capable of receiving digital signals from the embodiment of the receiver portion of the device. And intern, providing electrical and, or digital communication. Such as between the power source and the function enabling component and, or combinations of components comprising the embodiment of the device. Relative to the specific portion of the controls remotely activated. On the remotely employable set of controls for the embodiment of the device. Of which, in addition to all controls, capable of transmitting all signals, for the remote activation, of and variable control all servos, and servo motors. Such as #50-A. and #50-B. Included in FIG. 9. Through FIG. 11. And #22 Included in this drawing of an embodiment of the device. Along with all of the functions and service providing abilities providable by the embodiment of the device. Enabled by same.

[0062] Of which may additionally include controls capable of transmitting signals to the receiving antenna and receiver portion of an embodiment of the device and thus, also to the switching and control component portions, of the embodiment thereof. Capable of distributing power and or digital communication.

[0063] Relative to the portions of the controls for same, selectively employed, by the soldier, remotely employing the embodiment of the device. By means of the set of controls and military display device provided and employed in conjunction with the embodiment of the device. (Also not

included in this drawing, of an embodiment of the device. However, a couple of optionally employable embodiments thereof, are included in. FIGS. 12-A. And FIG. 12-B. Of which includes controls capable of additionally enabling the remote and variable controllability, of any number of additional function, enabling and service providing component portions, of an embodiment of the device.

[0064] Such as but may not limited to. The soldier's ability to variably adjust the digital video image sensors level of magnification. As well as the ability to remotely turn optionally employable portions of the embodiment of the device on and off. Such as the component portions capable of providing an embodiment of the device with night viewing and, or thermal optic viewing abilities. as well as controls capable of remotely activating switching components, portions of the embodiment of the device. Capable of turning the infrared illuminator portion of this embodiment of the device on and off as well as to control the illuminated intensity and beam width of same. Or capable of controlling the activation of the composition of the electronic component portions, of other embodiments of the device capable of providing the digital video image sensor portions thereof, with thermal optic viewing abilities.

[0065] This as well as for the remote activation and, or variable adjustability of any additional function enabling component portions of an embodiment of the device. Made as remotely controllable by means of the set of controls provided for the embodiment of the device and viewable in the screen of the military digital display device. Employed in conjunction with the embodiment of same. Of which, examples of employable embodiments thereof, are included in and described in further detail in some of the additional drawings and descriptions of other embodiments of the device and other portions thereof. Along with the very specific modes. In which the composition of all mechanical electrical and electronic components, comprising embodiments of the device. Included in this application. Achieve their abilities to perform all of the unique and extremely advantageous attributes along with all of the needful and far superior service providing abilities. Providable by embodiments of the device. Included in this application. In this example of an embodiment thereof.

[0066] However, all embodiments of the device. Included in this application, were also devised and engineered in the manner in which they were. To provide embodiments of the device with the optional ability to simultaneously employ the same digital video image sensor #2-A. portion of the embodiment of the device, to provide all operating visuals for the military vehicle, equipment or military robotics in which its employed, as well. And thus. Not only provide, both with a full vertical as well as a full 360-degree range of view for all operating visuals for both, along with the plurality, of additional, unique function enabling service providing abilities, performable by the digital video image sensor portion of the device, enabled by means of the unique modes, in which the digital video image sensor, is employed in embodiments of them and the combination of unique and extremely effective service providing abilities that are additionally provided by embodiments of the device, to both.

[0067] But all were also devised and engineered in the ways in which they were. To provide embodiments of them, with the ability to even further expand, the levels and range of needful service providing abilities, providable by embodiments of the device, even much further, when installed on a remotely controllable type of military equipment, or robotics, by additionally enabling. Not only the remotely employable controls, for the embodiment of the device to be easily included as a portion of the remotely employable controls, for the military equipment, or military robotics, in which its employed, or vice versa. But primarily to enable the employability of the digital video image sensor portion of the device employed on same, with the ability to provide all operating visuals for both. Thus, not only providing the soldier or squad, remotely employing same, with the simultaneous provision of all operating visuals, along with the remote employability, of all of the radically advanced, surveilling and service providing abilities provided by the embodiment of the device, in conjunction with all service providing abilities providable by the military

equipment, or military robotics. In which the embodiment of the device is employed. Thus, providing all providable by both, as a whole. Not only in an extremely operationally efficient manner. But also, by means of a very minimal number of apparatuses and components needfully employed to enable the provision of all services and abilities providable by both. While doing so. In this example of an embodiment thereof

[0068] FIG. 2. Includes a drawing of an embodiment of two brackets, #44-B. and #44C. capable of providing a mechanical connection between the outer side portions, of the vertically adjustable portion of the embodiment of the device, and mechanically fastened to same. by means of mechanical fasteners. #28-D. Shaft #45-B. Is an integral portion of brackets. #44-B. And of which Shaft #45-B. would be mechanically connected to a crown gear portion of an embodiment of the device #48-B. (Of which is not included in this drawing. But an embodiment of crown gear #48-B. Along with the basic composition of all additional electrical and mechanical components, capable of performing a full range of vertical angular adjustability to an embodiment of the device. would be housed within the inner housing portions of #25-A. And of which are viewable in the drawings and described in detail in the descriptions provided for. FIG. 9 and FIG. 11.) In this example of an embodiment thereof.

[0069] FIG. 3. Simply includes a drawing of an embodiment of a remotely operable military surveilling and service providing apparatus, as it would appear with the rotatably adjustable base portion of this device #27. Capable of providing horizontal adjustability to the embodiment of the device. Rotated around and drawn as if this embodiment of the device was inverted and mounted beneath an overhang, or under another portion of the military vehicle, equipment, or military robotics. In which this embodiment of the device is employed. The basic function enabling composition of components actually housed within the housing portion. #1-B. of this embodiment of the digital video image sensor sighting portion of this embodiment of the device. Would actually provide only slight variations of the same basic function enabling service providing abilities, as the embodiment thereof. Included in FIG. 1. However, the actual digital video image sensor sighting portion of the embodiment thereof, included in. FIG. 1. Includes several slight differences relative to, the embodiment thereof. Included in this drawing. However, the basic composition of function enabling components housed within this embodiment thereof. Is included in an overhead drawing and detailed description of this embodiment thereof in. FIG. 6. In this example of an embodiment thereof.

[0070] FIG. 4. Includes a drawing of a fully self-contained embodiment of a remotely positionable surveilling and target acquiring device. Of which unlike the embodiments thereof. Included in FIG. 3 and FIG. 1. This embodiment of the device would include embodiments of, all of the basic function enabling composition of components, housed within the actual housing portion of this embodiment of a digital video image sensor housing portion. #1-C. Of this embodiment of the device. To enable the remote employability of all functions and service providing abilities providable by this embodiment of a remotely positionable surveilling and sighting device. By means of the remotely employable set of controls and digital display device for same (Not included in this drawing). The greatest portion of all of the basic function enabling composition of components housed within this embodiment of the digital video image sensor sighting portion of the device. Is included in the drawing of an overhead view of the embodiment of the digital video image sensor sighting portion of the device, included in. FIG. 6. Of which includes its own power supply, capable of providing all power requirements for this embodiment of the device. Comprising at least one battery, housed within a battery compartment #32. Portion of this embodiment of the device,

[0071] This as well as a digital video image sensor, #2-B. primarily with a wide un-magnified yet zoomable field of view with the lens portion thereof, centered directly behind and adjustable in conjunction with reticle, #14. Although of course, any type of actual reticle, would be employable by this portion of any embodiment of the device. Included in this application. This type of reticle,

was simply included, in this drawing, as well as other drawings and descriptions of embodiments of this portion of the device. Included in this application. As an example of a reticle. However, embodiments of the device would also be equipable with or provided with any other types of actual reticles as well as the adjustability thereof relative to or in conjunction with the center of the viewpoint of the digital video sensor portion of the embodiment of the device.

[0072] This drawing of an embodiment of the device, also includes an embodiment of an omnidirectional transmitting antenna #23-A. Capable of receiving all digital video images from a transmitter portion of the device. (Not included in this drawing of an embodiment thereof) captured by the digital video image sensor portion of this embodiment of the device and transmitting same, to a receiving antenna and receiver portion of one of many types of remotely employable military digital display devices. (Also not included in this drawing.) However, of which would be capable of both receiving and displaying all digital video images transmitted to same by the transmitter and omnidirectional transmitting antenna #23-A. Portion of an embodiment of the device.

[0073] This embodiment of the device, would also be equipped with a digital receiver, (Also not included in this drawing of an embodiment thereof) As well as an omnidirectional receiving antenna #24-A. Capable of receiving digital signals from a transmitter and transmitting antenna portion of a remotely employable set of controls. For this embodiment of the device. (Also not included in this drawing). This embodiment of the device would also include switching and controlling components (also not viewable in this drawing) However of which are included in the drawing and description, provided for FIG. 5. Of which, would be capable of receiving digital signals from the receiver portion of the device and providing electrical communication between the power supply and all function enabling component portions of this embodiment of the device needful of same, relative to the portions of the remotely employable controls for the embodiment of the device, selectively employed by the soldier, remotely employing same.

[0074] However, this drawing of an embodiment thereof. Also includes a manually employable reticle selector #21. As well as switch #40. Capable of providing electrical communication between the power supply and the rest of the composition of electrical components. Comprising this embodiment thereof. For the powering up of the embodiment of the device. This as well as #41. Also simply comprising a switch, capable of completing a circuit and providing electrical communication between the power supply and the thermal optic, enabling components, for the digital video image sensor portion of this embodiment of the device. (Also, not viewable in this drawing of an example of an embodiment thereof.

[0075] There's also, another drawing of an embodiment of the digital video image sensor housing portion of a fully self-contained embodiment of the device, equipped with embodiments of the greatest portion of the same basic composition of function enabling components employed within the digital video image sensor housing portion. Of this example of an embodiment of the device. This embodiment, of a digital video image sensor housing portion of an embodiment of the device also includes a power supply, capable of providing all power requirements for the embodiment of the device. Also, included within the housing portion of this embodiment of a digital video image sensor housing portion of an embodiment of the device, includes at least one battery, capable of providing all the power requirements, for the composition of all components, capable of enabling the functionality and remote controllability of all functions and services, providable by the embodiment of the device, as a whole. #32. Comprises a battery compartment portion of this embodiment thereof capable of housing said power supply for this embodiment of the device. #37. Simply comprises a door, capable of providing access to the battery, or batteries housed within this embodiment of a battery compartment. Portion of an embodiment of the device.

[0076] FIG. 5. Includes a drawing of an overhead view of an embodiment of a digital video image sensor housing portion, of an embodiment of the device. Included in this application. This embodiment thereof, is drawn as if the upper housing portion, of this portion of an embodiment of the device, was transparent, to enable the easy viewing, of this example of a large portion of the

composition of the basic function enabling components. Optionally employable within this portion of an embodiment thereof. The basic composition of function enabling components, housed within and stationed upon. This embodiment of the digital video image sensor housing portion, of an embodiment of the device that are actually viewable in this drawing of an embodiment thereof. Firstly, includes, its own power supply. capable of providing all power requirements for the embodiment of the device. Comprising at least one battery. #38. Capable of providing all the power requirements, for the composition of all components, enabling the functionality and remote controllability of the greatest portion of all functions and services providable by the embodiment of the device, as a whole, housed within a battery compartment #32. Portion of this embodiment thereof.

[0077] #37. Included in this drawing, simply comprises a door, capable of providing access to the battery or batteries housed within this embodiment of a battery compartment. Portion of an embodiment of the device. #2-B. Included in this drawing of an embodiment thereof is a digital video image sensor. Of which all embodiments of this portion of the device are, equipped with and employ in conjunction with, additional component portions of all embodiments of the device. Of which, not only provides the digital video image sensor portion of this embodiment of the device, with the ability to simultaneously perform a plurality of basic function enabling service providing abilities, performable by this embodiment of the device, at the same time.

[0078] But the levels and range of the service providing abilities, attainable by the digital video image sensor portion of this embodiment of this portion of the device, are even further amplified and brought to all new levels. By means of several additional unique features and attributes also exclusive unto the embodiments thereof. Included in the embodiments thereof included in FIG. 3. through FIG. 6. Such as, but far from limited to, the unique modes in which the digital video image sensor housing portions thereof, are configured, as well as the modes in which the digital video image sensor as well as several components are stationed and employed in conjunction with same.

[0079] The first of these attributes. Exclusive to the digital video image sensor housing portions thereof. Includes the incorporation and positional orientation of side viewing windows. #10-C. Stationed in the forward side portions of each side of the digital video image sensor housing portions of the device. Of which include aft portions thereof position within same. In this example of an embodiment thereof. laterally parallel from the base of the of the lens #5-B. portion of the digital video image sensor portion of the device and extending therefrom, all of the way to the forward outer end portions of the digital video image sensor housing portions of the embodiment of the device where the forward end portions of the digital video image sensors side viewing windows, are seamlessly connected to. Or produced as an integral portion of, the digital video image sensors forward viewing window #10-B. Of which, along with the unique modes in which the digital video image sensor and reticle portions of the device are disposed and employed within this embodiment thereof. Additionally provides the lens portion #5-B. of the digital video image sensor. #2-B. With a totally unobstructed field of view. Many times, greater than other types of sighting devices, could possibly achieve. But also provides the simultaneous provision of this embodiment of a sighted image of a reticle #14. Centered in all of the surveilled digital video images captured and provided, by the digital video image sensor #2-B. Portion of the device. With this embodiment thereof, also comprising the reflected image of a tiny light beam #20. In perfect alignment with the center portion of this embodiment of the lens #5-B. portion of the digital video image sensor #2-B. In this embodiment thereof and provided by means of a reticle projector. #19. Included in this embodiment thereof.

[0080] This drawing of an embodiment of this portion of the device, also includes a large portion of employable embodiments, of the basic function enabling components capable of transmitting all digital video images, captured by the digital video image sensor. #2-B. Portion of an embodiment of the device. Either, to the military digital display device. Or to the digital display portion of a remotely employable set of controls, provided for an embodiment of the device, remotely

employed, by the soldier or squad employing same, in conjunction with the embodiment of the device and capable of enabling the remote viewing of all digital video images captured, by the digital video image sensor portion of the embodiment of the device.

[0081] Also stationed within the digital video image sensor housing portion of this embodiment thereof, includes an embodiment of a digital video image transmitter. #3-B. Capable of receiving digital video images captured by the digital video image sensor #2-B. And transmitting same. either by means of direct wire or fiber-optic, digital video image communication through cable #8. Included in this drawing of an embodiment thereof and, or by means of wireless digital video image transmission. By means of an omnidirectional transmitting antenna. Also included in this drawing of an embodiment thereof as #23. Capable of transmitting all digital video images, captured by the digital video image sensor portion of this embodiment thereof and received from a wireless digital video transmitter. Also represented as. #3-B. Included in this drawing of this portion of an embodiment of the device, and transmitting same, to a receiving antenna and receiver portion of a remotely employable military digital display, or digital display device. Or to the digital display portion of the set of remotely employable set of controls, for an embodiment of the device, remotely employed, in conjunction with the embodiment of the device. (Not included in this drawing.) However, a drawing of a couple of embodiments thereof, are Included in FIGS. 12-A and FIG. 12-B. This portion of this embodiment of the device, also includes a receiver. #7. Comprising either, an embodiment thereof. Capable of wirelessly receiving digital signals from a receiving antenna. Portion of the device (Not viewable in this drawing of an embodiment thereof.) Capable of receiving digital signals transmitted from a transmitter and transmitting antenna portion of an embodiment of the remotely employable set of controls, provided for and employed in conjunction with, an embodiment of the device. Such as employed in an embodiment of the device, equipped for employment of same on a remotely controllable type of military equipment or military robotics. Or optionally with #7. Comprising a receiver, capable of receiving digital signals, by means of direct wire or fiber optic, digital communication from a direct wire or fiber optic transmitter portion of an embodiment of the remotely employable controls, provided for the embodiment of the device, capable of transmitting digital signals from same, to the embodiment of this portion of an embodiment of the device, by means of direct wire and, or fiber-optic, digital video communication. And thus, enabling the remote employability and the provision of all services and abilities providable by the embodiment of the device, by the soldier employing same. by means of direct wire or fiber-optic, digital communication. Such as in the employment of an embodiment of the device, equipped for the employment of same on an armored military vehicle, or other, and the remote employability of same along with the provision of all needful service providing abilities providable by the embodiment of the device availed to the soldier, by means of his employment of an embodiment a digital display and set of controls provided for same. Such as the embodiments thereof. Included in FIG. 13. Remotely stationed inside of same.

[0082] #39. Included in this drawing of an embodiment thereof. Comprises an embodiment of a switching and power distribution module. Comprising a plurality of electronic components on a circuit board, capable of receiving digital signals from the embodiment of the receiver. #7. And intern, providing electrical communication between the power source and the function enabling component and, or combinations of components comprising the embodiment of the device. Relative to the specific portion of the controls remotely activated. On the remotely employable set of controls, provided for the embodiment of the device. Relative to the needful services and abilities providable by the embodiment of the device remotely operated by the soldier employing same.

[0083] The remotely employable controls. Employed in conjunction with, the embodiment of the device employed. Along with the embodiment of a switching and power distribution module #39. Also provided for the embodiment of them. Would all at least provide the soldier employing the embodiment of the device, with the ability to remotely activate and variably control all servos, and

servo motors. Such as #22, #50-A. and #50-B. Included in FIGS. 1, FIG. 9. FIG. 10-A, and FIG. 10-B As well as FIG. 11. And thus, with the ability to remotely employ all functions and service providing abilities performable by the embodiment of the device, enabled by same. However, most switching and power distribution modules as well as most embodiments of the controls provided as employable in conjunction with the embodiment of the device. Would also, provide the soldier with the ability to remotely activate and control any number of additional functions and service providing abilities, providable by the embodiment of the device employed. Such as, but far from limited to. The soldier's ability to variably adjust the digital video image sensors level of magnification. As well as the ability to remotely turn optionally employable portions of the embodiment of the device on and off. Such as the components capable of providing embodiment of the device with night viewing or thermal optic viewing abilities. Of which is provided by means of the composition of electronic components. Included on the two circuit boards #16. In this embodiment of this portion of the device. Of which would be capable of providing the digital video image sensor portion of this embodiment thereof, with thermal optic viewing abilities. Or the remote provision and control, of any other service of function enabling components or the variable adjustability of same. By means of the set of controls provided for the embodiment of the device and thus also viewable in the screen of the digital display or digital display device, provided for the embodiment of device, remotely employed, by the soldier employing same. In this as well as all embodiments of the device included in this application. In this example of this portion of an embodiment of the device.

[0084] FIG. 6. Includes a drawing of an overhead view, of one of many optional embodiment variations, of the digital video image sensor housing portion of an embodiment of the device. This drawing of an embodiment thereof. Like the embodiment of this portion of an embodiment of the device. Included in FIG. 5. Was also drawn as if the housing portion. 1-B. Thereof, was transparent, to provide the composition of components, housed within this embodiment of this portion of the device, with the ability to be easily viewed. Notice that, this drawing of an embodiment of a digital video image sensor housing portion of an embodiment of the device, share all things common, with the forward portion of the digital video image sensor housing portion of the embodiment thereof. Included in FIG. 5. But includes only a portion of the basic function enabling composition of components, housed within this embodiment thereof. As is Included in. FIG. 5. However, the composition of all components stationed within the forward portion of the digital video image sensor housing portion of this embodiment of this portion of the device, would be employed in the same basic manner and perform the same basic function enabling contributions, to the functionality of the device as is performed by the same portion of the basic function enabling components. Included in the embodiment thereof, included in FIG. 5. And of which were simply drawn in the same basic manner, as those. Included in FIG. 5. And as a result, thereof. All component descriptions, as well as the descriptions of the function enabling contributions. Relative to same. Included in FIG. 5. Would also be applicable to the same basic component portions of the device. Included in this drawing of an embodiment thereof.

[0085] This drawing of an embodiment of a digital video image sensor housing portion of an embodiment of the device. As well as the basic composition of components housed within same. Was simply added, to include a drawing of one of many optionally employable embodiment variations, of this portion of an embodiment of the device. Although the only real differences between this embodiment thereof and the embodiment thereof. Included in. FIG. 5. Is in the reduced number of the actual function enabling components, that are actually stationed within the actual digital video image sensor housing portion. Of this embodiment of this portion of the device. Of which In of which would simply stationed either on or within other portions of the embodiment of the device it self or the military vehicle, equipment or military robotics, in which the embodiment of the device is employed. Of which could and may include any number of the basic function enabling, component portions of an embodiment of the device, that doesn't actually need

to be housed within the digital video sensor housing portion of a particular embodiment of the device, to enable the composition of all electrical and electronic components, comprising the particular embodiment of the device with the ability to perform all of services and abilities performable by same.

[0086] Such as but not necessarily limited to. All additional electrical, and electronic, digital, video image transmitting, receiving and switching component portions thereof or the actual power supply for the particular embodiment of the device. Regardless of the actual distance in which the various portions thereof are stationed, Relative to each other. Of which would be accomplished primarily by means of direct wire and, or fiber optic communication. And transferred between same and all separately mounted portions thereof by means of cable **8-B**. In this example of an embodiment thereof.

[0087] Most embodiments of the device. Included in this application. Also acquire the greatest portion of their attributes and far superior service providing abilities, performable by embodiments of them. In very similar ways and do so by means of a composition and employment of various portions of the same basic function enabling mechanical, electrical and electronic components and combinations thereof, in conjunction with each other. In the same or very similar basic foundational modes, common to all.

[0088] Of which enables various portions thereof, with the ability to further employ each other's basic function enabling contributions, in conjunction with their own. In very unique ways, that benefits and expands, the levels and range of each other's service providing abilities in ways that literally provides each, with the ability to perform a plurality of basic function enabling contributions, to the overall functionality and range of service providing abilities. providable by embodiments of the device, as a whole. While doing so.

[0089] Such as the unique modes in which the digital video image sensor. Along with the composition of all additional components, needfully employed within and in conjunction with embodiments of his portion of embodiments, of the device. Included in this application, are disposed and employed within embodiments of them. Of which provides this portion of this embodiment of the device. Not only with the ability to simultaneously perform a plurality of service providing abilities. Impossibly performable by any single apparatus previously devised and do so, by means of a small percentage of the actual composition components and complex systems needfully employed by anything previously devised, to possibly do so. But the actual number of function enabling components, needfully housed within the digital video image sensor housing portions of embodiments of the device device, included in FIGS. 7-A. and FIG. 7-C. Are even further reduced. In that, unlike the embodiment thereof, included in this drawing as well as most other embodiments of this portion of the device. Included in this application. Embodiments thereof included in FIGS. 7-A and 7-C. Are neither equipped with components capable of producing a reticle, or the adjustability thereof. In that both would be provided for them, simply by means of enabling software, or instructions included in enabling software embedded into the operating system of the computerized military display device, equipped to be employed in conjunction with embodiments of them. Of which along with the modes in which the lens portions of the digital video sensors are stationed and employed on the outer portions of the digital video image sensor housing portions of the embodiments thereof. Included in FIGS. 7-A. FIG. 7-C. And FIG. 7-D. Also provides embodiments of them with additional attributes. exclusive unto themselves.

[0090] Relative to the embodiment of this portion of the device, included in this drawing, as well as many of the other embodiments thereof included in this application. Of which also possess several advantageous attributes. Exclusive unto themselves as well. In addition to the advanced level and range of needful service providing abilities, providable by all embodiments of them. Included in this application.

[0091] Of which, a portion thereof is attributable, in part, to several unique function enabling features and attributes. Exclusive to the digital video image sensor housing **#1-B**. portions of the

embodiment of this portion of the device. Included in this drawing, as well as the embodiment thereof. Included in FIG. 5, and several other embodiments thereof. Included in this application. Of which were devised and engineered, along with the mode in which the digital video image sensor, along with the composition of all additional components, needfully housed within this portion of the embodiments thereof, are employed relative to and in conjunction with each other.

[0092] Of which among other things, includes the center of the lens portion of the digital video image sensor stationed within the digital video image sensor housing portion of them, directly behind and sight adjustable to, or in conjunction with the reticle. In this as well as many other embodiments of this portion of the device. Included in this application. Of which, not only provides the digital video image sensor portions of the device, with the ability to simultaneously perform a plurality of far superior sighting and surveilling abilities, along with the ability to do so by means of a small portion of the function enabling composition of components and complex electronic systems needfully employed in and by any apparatus or combination of apparatuses previously devised.

[0093] But the levels and range of the service providing abilities, providable by embodiments of this portion of the device. Included in this drawing, as well as several other drawings of embodiments thereof. Included in this application. Are brought to even far greater levels, by means of several features and attributes. Also exclusive to this as well as many other embodiments of the digital video image sensor housing portions thereof. Included in this application, along with the modes in which the composition of many of the function enabling components are employed within same. Such as the mode in which the digital video image sensors viewing windows or window portions. Included in this drawing of an embodiment thereof, are stationed and employed within the digital video image sensor housing portion of this, as well as well as several other embodiments thereof. Included in this application.

[0094] Of which includes, both side viewing window portions #10-C. As well as a forward viewing window #10-B. Portion, of the digital video image sensor housing portions of the embodiments of them. Of which would be either seamlessly put together, or produced as a single window. In this as well as in most other embodiments, of this portion of the device. Included in this application. Of which are stationed within, or on the embodiments of the digital video image sensor housing portions of them. In a manner that provides of the lens portion #5-B. of the digital video image sensor #2-B. Portions of the device stationed within the embodiments of them, with a totally unobstructed field view, many times that of any other sighting or surveilling device previously devised.

[0095] With the exception of the embodiments thereof. Included in FIGS. 7-A. and FIG. 7-B. Of which would also possess fields of view. limited only by the UN-magnified field of view of the digital video image sensors employed within embodiments of them. Of which all, not only provides these portions of the embodiments of the device with fields of view, many times, greater than any other type of sighting or surveilling device, previously devised and provides same to the users of them, in a manner that's. Not only instantly magnifyable but variably magnifyable in the screens of the digital display device portion of the control's provided for and employed in conjunction with embodiments of them

[0096] But the range of needful services and advanced surveilling abilities, providable by embodiments of them, and thus the military vehicle, equipment or military robotics, as well as the soldier or squad employing same, are even much further multiplied by means of embodiments of the composition of additional mechanical electrical and electronic component portions of the embodiments of the device. Included in this application, as a whole. Such as the embodiments thereof, included in FIG. 1 as well as those included in FIG. 3, and FIG. 4, and employed as a single unit on a military vehicle, equipment or military robotics.

[0097] Of which additionally provides the soldier or squad remotely employing the embodiment thereof with a full vertical, as well as a full 360 degrees of horizontal employ-ability of all

surveilling and service providing abilities. Providable by embodiments of the device, relative to the military vehicle, equipment or military robotics in which its employed, from positions of full protective cover and totally out of the possible line of incoming fire, while doing so. Of which will doubtless result in a substantial reduction in the future number of U.S. casualties.

[0098] But embodiments of the device were also devised and engineered, much in the ways in which they were. To provide the digital video image sensor portion of embodiments of the device. Not only with the ability to provide all sighting, surveilling and operating visuals for the embodiment of the device itself. But also with the ability to simultaneously employ the same digital video image sensor portion of the embodiment of the device, to provide all operating visuals for the military vehicle, equipment or military robotics in which the embodiment of the device is employed, While doing so, by means of the same controls and military display device, provided for and employed in conjunction with the embodiment of the device. In this as well as all other examples of embodiments thereof. Included in this application.

[0099] FIG. 7-A. Through FIG. 7-D. Include overhead drawings of the bottom portions of additional, optionally employable embodiments of the digital video image sensor housings along with the composition of components employable and housed within these particular embodiments of the digital video sensor housing portions of the device. Of which, for easy reference purposes were drawn as if these embodiments thereof were employing many of the same basic types and variations of many of the basic function enabling composition of components as are employed within the embodiments thereof included in FIG. 5 and FIG. 6. As well as several other embodiments thereof. Included in this application and as a result, thereof. All component descriptions, relative to the same components. Included in FIG. 5 and FIG. 6. Would also be applicable to the same basic component portions of the device. Included in the drawings of these embodiments thereof. These embodiments thereof, also share many additional things common with the embodiments thereof included in FIG. 5 and FIG. 6. as well.

[0100] However, the embodiments thereof. Included in the drawings of FIGS. 7-A. FIG. 7-C. And FIG. 7-D. Actually possess several distinct advantages, to all of the other embodiments of this portion of the device. Included in this application. Of which, in part is provided by means of the mode in which the actual lens portions #5-B. of the digital video image sensors. #2-B. And 2-C. Are stationed and employed on and by the embodiments of these portions of the device. Included in the drawings of FIGS. 7-A. FIG. 7-C. And FIG. 7-D. Of which is accomplished, in part by means of the lens portions 5-B. Thereof actually being stationed on exterior portions of the device. Of which, not only provides these embodiments thereof with the unrestricted employment of any type of optically zoom-able digital video image sensor, without hindering their range, of external extend-ability, of optical telescopic lens, or field of view in any way. But also eliminates the possibility of any of the images captured by these embodiments thereof from being hindered, distorted or restricted, in any way, as they pass through the viewing window or windows provided in any of the other embodiments of the digital video image sensor housing portions of the device. Included in this application. However, notice there are also several additional distinctive differences between the embodiments thereof. Included in FIGS. 7-A. FIG. 7-C and FIG. 7-D. As well.

[0101] FIG. 7-A. Includes a drawing of an overhead view of the bottom portion of an embodiment of a digital video image sensor housing portion of an embodiment of the device. This drawing of an embodiment a digital video image sensor housing portion of an embodiment of the device, as well as the drawing of the multi digital video image sensor embodiment thereof. Included in FIG. 7-C. Also provide a clear view of the some of the pre-referenced differences possessed by these embodiments thereof. Such as, but far from limited to, the mode in which the telescopic lens #5-B. portions of the digital video image sensor portions thereof are actually stationed on exterior portions of the housings of these embodiments thereof. This drawing of this particular embodiment of a digital video image sensor housing portion of an embodiment of the device also provides a

clear view of a mounting portion of this embodiment thereof. #9. Stationed under the battery compartment. #32. Portion of this embodiment thereof, capable of providing this portion of this particular embodiment thereof, with its own power supply. This drawing of an embodiment thereof also includes manual switches #40. Capable of providing electrical communication between same and the digital video image sensor #2-B. And transmitter #3-B. Portions of this embodiment thereof, as well as switch #41. Capable of providing electrical communication between the power supply and the night viewing, or thermal optic viewing composition of components. #16. Provided for this embodiment thereof. However like all other embodiments of the device. Included in this application. The actual composition of components capable of providing electrical communication between the actual power source, provided for the particular embodiment of the device, capable of enabling the activation and control of these, as well as the activation and control of the composition of various other function enabling, electrical and electronic components, comprising embodiments of the device. Included in this application. Could and may be enabled and controlled. Both by means of the composition of wirelessly activated switching and control components provided, or portions of the particular embodiment of the device, as well as by means of the enabling software and instructions, provided and downloaded into the operating system of the dedicated computerized digital display device and controls provided for and employed in conjunction with the particular embodiment of the device.

[0102] FIG. 7-B. Simply includes a drawing of an embodiment of a removable inspection or access plate. #67. Providable for the embodiments thereof. Included in with FIGS. 7-A, FIG. 7-C and FIG. 7-D. With this embodiment thereof being mechanically fastenable to and removable from bottom portions of the component housing portions of these particular embodiments thereof by means of the removal mechanical fasteners #28-E. And a seal for same. That would also be provided for this embodiment thereof. Providing access to the composition of components housed within the inner housing portions of the embodiments thereof. Included in FIGS. 7-A, FIG. 7-C and FIG. 7-D. Simply to enable the easy assemblage, as well as easy access to same, in the possible case of repairs, or future maintenance needfully performed on same. In this example of an embodiment thereof.

[0103] FIG. 7-C. Also Includes a drawing of an overhead view of the bottom portion of an embodiment of this portion of the device. Of which is actually quite different to any of the other embodiments thereof. Included in this application. In that, in addition to the forward viewing digital video image sensor #2-B. Capable of capturing sighting and surveilling images for the embodiment of the device. This embodiment of the digital video sensor housing portion, of this embodiment thereof, is not only equipped with additional digital video image sensors. #2-C. and a transmitter or transmitters, #3-B Capable of capturing and transmitting additional fields of view.

[0104] Exclusive unto themselves. But embodiments of the device. Included in this drawing, would also be optionally employable in conjunction with an embodiment of the device. Included in FIG. 7-D. Of which comprises an embodiment of a rear viewing surveilling device, also capable of capturing and transmitting an additional field of view either wirelessly, by direct wire, or by means of fiber optic cable to an embodiment of a computerized dedicated digital display device and controls provided for and equipped to be employable in conjunction with embodiments of the device. Included in this drawing of an embodiment thereof. Of which, would also be optionally employable with any additional controls as well as any additional software and instructions downloaded into the operating system of same, capable of providing the soldier or squad employing an embodiment of the device. Not only with the ability to selectively or display and thus view any of of all the fields of view, provided by the embodiment of this portion of the device, full screen, along with the additional ability to simultaneously display and thus view any number of, or all fields of view provided to the display device, captured by any or all of the individual digital video image sensor portions of an embodiment of the device. Included in this drawing of an embodiment thereof.

[0105] But when employed in conjunction with an embodiment of the rear viewing surveilling device. Included in FIG. 7-D. Also cable of capturing and transmitting surveilled digital video images to an embodiment of the digital display device and set of control's. Provided for an embodiment of the portion of the device. Included in this drawing of an embodiment thereof.

[0106] Mounted either on a rear portion of an embodiment of the device itself. Or simply mounted on a rear portion of the military vehicle, equipment or military robotics in which the embodiment of the device equipped with both is employed.

[0107] Will provide any and all of our troops employing embodiments thereof. Not only with all of the desperately needed safety and control providing abilities, providable by all embodiments of the device. Included in this application. But will additionally provide this embodiment thereof with the ability to provide the soldier or squad employing embodiments thereof. With an ongoing full 360 degree view of their entire surroundings and all of the otherwise impending dangers therein 24-7. From safe positions of total protective cover. Simply by means of leaving the device on and monitoring the digital video screen portion of the actual composition of controls provided for the embodiment of the device, without having to rotate or adjust the viewing angle of the device as a whole, by means of the composition of wirelessly activated switching and control component portions of the controls, provided for, or portions of the particular embodiment of the multi digital video sensor embodiment of the device.

[0108] This drawing of an embodiment of this portion of the device, like the embodiment thereof. Included in FIG. 7-A. Also includes its own battery compartment #32. Of which is actually viewable in this drawing of an embodiment thereof. In that the bottom portion of a battery compartment #32. Is actually drawn as if same was transparent to enable the easy viewing of a power supply. Provided for this particular embodiment thereof. Of which includes

[0109] at least one battery #38. Housed within same. This well as an embodiment of an access panel for same #37. Capable of providing access to the power supply housed within this embodiment thereof. like the embodiment thereof included in FIG. 7-A. This particular embodiment thereof also includes manually employable switches. #40. Capable of providing electrical communication between the power supply and the plurality of digital video image sensors #2-B. And 2-C. And transmitter or transmitters #3-B. Also included in this portion of this embodiment thereof. Capable of transmitting digital video images captured by 2-B. And 2-C. To a digital video display device employed in conjunction with same. This as well as switch #41. Capable of providing this embodiment thereof with manually controllability of the electrical communication between the power supply and the night viewing, or thermal optic viewing composition of components. #16. Also provided in this example of an embodiment thereof.

[0110] FIG. 7-D. Includes a drawing of an overhead view of an embodiment of the bottom portion of a rear viewing digital video image sensor surveilling device. Of which was also drawn as if the actual housing portion of this embodiment thereof was transparent, to enable the easy viewing of the composition of components comprising this particular embodiment thereof. Embodiments of these rear viewing digital video image sensor surveilling devices would be employable in conjunction with any embodiment of the device included in this application. However this embodiment thereof was referenced in the description provided for. FIG. 7-C. Of which also included a description of the basic mode in which an embodiment thereof could be employed in conjunction with the embodiment of the portion of the device. Included in. FIG. 7-C.

[0111] This drawing of an embodiment of a rear viewing digital video image sensor surveillance device. like the embodiment of the portion of the device. Included in FIG. 7-A. and FIG. 7-C. Is also drawn as if it was equipped with its own power supply, housed within its own battery compartment #32. Of which may or may not actually be included in this, or any other embodiments of the rear viewing digital video image sensor surveilling device. Included in this application. Or any embodiments of the digital video sensor housing portions of any embodiments of the device, or the power supply provided for any embodiments of the device as a whole. Included in this

application, for that matter. In that the actual power supply provided for embodiments of the device in most military applications of embodiments of the device would be provided by a separate, or by the main power supply, provided for the military vehicle, equipment or military robotics in which the embodiment of the device is employed. However this drawing also includes an embodiment access panel #37. For the battery compartment #37. Included in this drawing of an embodiment thereof. Capable of providing access to the power supply housed within this drawing of an embodiment thereof. like the embodiments of the portions of the device. Included in FIG. 7-A. and FIG. 7-C. This particular drawing of an embodiment of a rear viewing digital video image sensor surveillance device also includes manually employable switches. #40. Capable of providing electrical communication between the power supply and the digital video image sensor #2-B. And transmitter #3-B. Of which, like the embodiment thereof. Included in. FIG. 7-A. and FIG. 7-C. Would also be capable of transmitting digital video images captured by 2-B. To the digital video display device employed in conjunction with same. This embodiment of a rear viewing surveillance device also includes a switch #41. Capable of providing this embodiment thereof with manually controllability of the electrical communication between the power supply and the night viewing, or thermal optic viewing composition of components. #16. Also provided in this example of an embodiment thereof. #67. Simply includes a mounting base portion of this embodiment of a rear viewing digital video image sensor surveillance device. In this drawing of an embodiment thereof. [0112] FIG. 8. Includes a drawing of the same embodiment of the portions of the device. As is included in. FIG. 1. FIG. 3. And FIG. 4. Capable of providing the soldier employing same both with the full vertical angular adjustability as well as the full 360 degrees of horizontal angular adjustability of the embodiment of the device. Relative to the military vehicle, equipment, or military robotics. In which the embodiment of the device is employed. Of which simply consists of an embodiment of a remotely controllable gimbal bracket portion of the device. Comprising both of. #25. Of which is the portion of the device capable of remotely positioning the embodiment of the device, into a full range of vertical angles relative to the military vehicle, equipment, or military robotics in which the embodiment of a mounting base #44-A. Portion of this embodiment of this portion of the device is mounted on. Of which is indicated by means of the curved arrow #35 included in FIG. 1. #27. Includes a portion of, or is integrally connected to #25. And of which is the portion of the device capable of remotely positioning the embodiment, of the device into a full 360-degree range of horizontal angles of adjustability relative to the military vehicle, equipment or military robotics in which this embodiment of the device would be mounted to. [0113] Of which is indicated by means of the curved arrow #34. Also included in the drawing of FIG. 1. #44-B. Includes two mounting plate portions of the vertical and horizontally adjustable portion of an embodiment of the device. Included in FIG. 2. Of which, both include a shaft #45-B. Only partially viewable in this drawing of an embodiment thereof. However, one of the shaft portions of, one of the mounting plates. #44-B. Comprises a mechanical connection to the crown gear portion of the eternal composition of mechanical components housed within this embodiment of #25. (Not viewable in this drawing but viewable in the drawings provided for FIG. 9 and FIG. 11.) #8. Comprises one or more wires and, or tiny fiber-optic cables, capable of passing through all needful portions of this embodiment of this portion of the device to provide all needful electrical and, or direct wire or fiber-optic communication between the digital video image sensor housing portion of the device, as well as to servos and servo motors, and of which also include #8 passing all of the way through this embodiment of this portion of the device as well as various other embodiments of the device. Included in this application, that are employed on a type of military equipment, vehicle, or other military applications wherein the actual power supply for the embodiment of the device and, or other controlling component portions thereof are stationed on the actual type of military vehicle, equipment or military robotics. In which the embodiment of the device is employed and wherein a form of either direct wire or fiber-optic communication can be employed instead of wireless communication between the embodiment of the device and the

controls for same. Of which is not only common. But is also always employed in military applications, wherein any form or amount of wireless communication between any type of military equipment and the displays and or the controls for same. Not absolutely essential to the function enabling employability of same, can be eliminated and substituted by means of direct wire or fiberoptic communication, for security purposes. This simply due, both to the electromagnetic signature, unavoidably omitted during wireless transmission and communication between electronic components and devices of all kinds. Of which, not only stands the risk of, being detected and located. But even includes the possibility of being jammed, or even intercepted by the enemy. In this example of an embodiment thereof.

[0114] FIG. 9. Includes a drawing of a blown-up view, of the same embodiment of the portions of the remotely positionable sighting and surveilling device, as is included in FIG. 8. Of which is also drawn as if the outer housing portions of #25. As well as the outer housing portions of. #27. Were also transparent, to allow the composition of electrical and mechanical components housed within this embodiment thereof, to be easily viewed. However, this drawing thereof is simply provided as one example of many modes in which this portion of the device could be engineered to possess the abilities to provide the function enabling services and abilities needfully performed by this portion of the device and the applicant is simply offering this electrical and mechanical composition of components features and attributes as an example of an embodiment thereof. Capable of doing so.

[0115] #45-A. Includes a shaft portion of the mounting base #44-A. Of which this embodiment thereof shows same as being mechanically fasten able to the military equipment or vehicle in which its employed, by means of the plurality of mechanical fasteners #28-A, however other embodiments of shaft 45-A. could and may simply be made as an integral portion of the military equipment or vehicle in which its employed. #47. is simply an embodiment of a friction washer stationed between the first housing portion of this embodiment of #27. Of which may or may not actually be includes in any embodiments thereof. This shaft intern extends through an opening in the center portion crown gear #48-A. of which is comprises a mechanical connection to an end portion of shaft #45-A. By means of set screws #43. In this example of an embodiment thereof. The second housing portion of 27. In this embodiment thereof. Of which would include the reversible motor. #49-A. Capable of rotating and a worm gear, portion of same #50-A. Capable of rotating the crown gear #48-A. connected to shaft #45-A. connected to or made as an integral portion of the military vehicle, equipment or robotics, in which the embodiment of the device is employed. And thus rotation of the entire device horizontally, to a plurality of angles both clockwise and counterclockwise relative to the military vehicle equipment or robotics, in which the embodiment of the device is mounted on when fully assembled, is achieved, by means of the remote activation of motor #49-A. In this example of an embodiment thereof.

[0116] This drawing also includes a blown-up view of the first and second housing portions of this embodiment of 25. Of which, have been taken apart by means of the removal of a plurality of mechanical fasteners #28-C. in this embodiment thereof. Of which simply includes the same basic composition of mechanical and electrical components, as is housed within #27. And that are included in FIG. 10. With the exception being the composition of components housed within #25. are provided and capable of rotating shaft. #45-B. Both clockwise and counterclockwise Of which is an integral portion of one of the mounting plates. #44-B. capable of comprising an attachment to and providing a full range of vertical adjustability to this embodiment of the device relative to the military equipment or vehicle, in which. This embodiment of the device is mounted to. Notice that the second mounting plate #44-B. portion of this embodiment of the device also includes an integral shaft portion of same. Of which is housed within an inset portion on the opposite side, or second portion of #25. Capable of providing a stable mounting platform for this embodiment of a second mounting plate #44-B. While additionally allowing same to freely rotate parallelly relative to the first mounting plate #44-B. Therein. In this example of an embodiment of this portion of an embodiment the device.

[0117] FIG. 10-A. Includes an overhead view of the same embodiment of the first housing portion of #27. As is included in FIG. 9, and FIG. 12. As if it had been removed from the second housing portion of #27. To provide a clear view of the composition of components, that would remain within this embodiment of the first housing portion of #27. Of which includes an end portion of shaft #45-A. Of which is a portion of an embodiment of a mounting plate #44-A. Of which includes shaft #45-A extending through an opening in the first housing portion of #27. And additionally extending through an opening in the center portion of an embodiment of a crown gear #48-A. Of which, This embodiment of #48-A. Comprises a mechanical connection to an end portion of shaft #45-A. By means of set screws #43. Threaded through flange portions of crown gear #48-A. and into inset portions of shaft. #45-A. In this example of embodiment thereof.

[0118] FIG. 10-B. Includes a drawing of an overhead view of the same embodiment of the second housing portion of #27. As is included in FIG. 9 and FIG. 11. Of which would be an integral portion of #25. In this embodiment thereof. Of which is drawn as if same, had been removed from the first housing portion of #27. To provide a clear view of the composition of components, that would remain within this embodiment of the second housing portion of #27. Of which firstly includes cable #8. Capable of providing electrical communication between the switching components and power source portions of an embodiment of the device. Including the embodiment of a tiny reversible motor #, 49-A. Included in this drawing of an embodiment thereof. Of which includes a worm gear portion of same #50-A. capable of meshing with and rotating the crown gear #48-A. (As seen in FIG. 10-A.) Once re assembled, and capable of rotating the second housing portion of #27. Included in this drawing of an embodiment thereof. Relative to crown gear #48-A. Included in FIG. 10-A. Of which is mechanically fastened to the shaft, portion of the mounting plate portion of this embodiment of the device. As seen in FIG. 9. And FIG. 11. By means of the set screws #43. (Also viewable in seen in FIG. 10-A.) Thus. once both the first and second housing portions of #27. Are reassembled. In this example of an embodiment thereof. And the teeth portions of worm gear #50-A. Mesh with the teeth of the crown gear. Included in FIG. 10-A. Angular adjustability of the entire embodiment of the device to a plurality of horizontal angles relative to the military vehicle, equipment or military robotics in which the embodiment of the device is mounted on. Is accomplished by means of the tiny reversible motor #49-A. Capable of rotating of the worm gear #50-A. clockwise and counterclockwise. In this example of an embodiment thereof.

[0119] FIG. 11. Includes a drawing of a profile view of the same portion of an embodiment of the device, as is included in FIG. 8. Through FIG. 10-B. With this drawing thereof including the composition of components, comprising this embodiment thereof, drawn as if same had been re-assembled. With this drawing also showing the first and second housing portions of #27. As well as the same first and second housing portions of #25. Of which contain the same basic function enabling composition of components. As is included in #27. This drawing is also drawn as if the outer housing portions of #25 and #27. Were transparent, so that a completely assembled view of the composition of components comprising these portions of an embodiment of the device could be easily viewed. Of which are also portions of the complete and fully assembled embodiments thereof. Included in FIG. 1. As well as in FIG. 3, and FIG. 4. In this example of an embodiment thereof.

[0120] FIG. 12-A. Includes a drawing of an embodiment of an employable set of controls and a digital display. Of which was draw in the manner in which this embodiment thereof was drawn. To represent an embodiment of a fairly simplistic set of digital controls and digital display device. Equipped with a composition of components, capable of maintaining all digital video image and digital signal control enabling communication, between same and an embodiment of the device. By means of direct wire and, or fiber optic digital signal transmitting and digital video image receiving components. Capable of receiving digital video images captured by the digital video image sensor portion, of an embodiment of the device, equipped with direct wire and, or fiberoptic, digital video image transmitting and digital signal receiving components.

[0121] Capable of transmitting digital video images captured by same and receiving digital signals, relative to choices made in the employment of the remotely employable switching and control component portions of this embodiment of a set of remotely employable controls, for an embodiment of the device. Equipped with direct wire and, or fiberoptic transmitting and receiving components.

[0122] Such as an embodiment of a set of controls and a display device. stationed within an armored military vehicle, or other, and capable of receiving and displaying digital video images and transmitting digital signals, by means of direct wire and, or fiberoptic communication, between same and an embodiment of the device. Stationed on an exterior portion of the vehicle or other. Thus, providing the safe provision and control of the many needful and far superior service providing abilities. Providable by the embodiment of the device, by means of the employment of the digital display device and controls, provided for the embodiment of the device. Remotely stationed inside a portion of the vehicle, or other. Additionally providing the soldier or squad safely stationed within same, with an ongoing visual of their entire surroundings and thus, with the ability to maintain positional awareness and very accurate knowledge of the true battlefield demographics, and of the present and otherwise unforeseeable, impending dangers residing within their surroundings, without having to come under fire from same. By means of their employment of the embodiment of a set of controls and a digital display device. Provided for the embodiment of the device, stationed within the armored military vehicle, or other.

[0123] The basic composition of controls. Included in this drawing of an embodiment of a set of controls. for an embodiment of the device, remotely employable by same. Actually, includes fewer switching and remotely employable function enabling controls, than could and would likely be included in the greatest portion of the embodiments of the remotely employable sets of controls, provided for embodiments of the device. Included in this application. However, the embodiment of the set of controls. Included in this drawing of an embodiment thereof. Firstly, includes portions of the controls capable of providing the soldier remotely employing an embodiment of the device, with the ability to remotely activate and variably control all servos, and servo motor portions of the device. Such as the portion of the set of controls. Included in this drawing of an embodiment thereof. Capable of providing the soldier remotely controlling an embodiment of the device with the ability to remotely control all vertical and horizontal angular positioning of the device, by means of control knob #55. Included in this drawing of an embodiment thereof. Of which includes a pivotable inner portion #57. of this embodiment thereof. Capable of remotely adjusting the vertical angle of the device, upward, by means of variable compressive pressure applied to a forward portion of #57. And the remote adjustability of the vertical angle of the device, downward, by means of variable compressive pressure applied to an aft portion of #57. Of this embodiment of control knob #55. This embodiment of control knob #55. Also includes a rotatable outer ring portion #61. Of which comprises the 360-degree horizontal angular adjustment portion of the controls, for the embodiment of the device. remotely controlled by same. Of which would be performed by means of the clockwise and, or counterclockwise rotation of the rotatable outer ring #61. Portion of this embodiment of control knob #55.

[0124] Also included in this drawing of an embodiment a remotely employable display device #59. And set of controls for an embodiment of the device. Includes #52. Capable of remotely activating and powering up the embodiment of the device, remotely employed and controlled by same. #53. included in this drawing of an embodiment thereof. Comprises of a compressible momentary switch, capable of remotely activating the trigger servo portion of the embodiment of the device. #54. Included in this drawing of an embodiment thereof. Comprises a switch capable of remote activating the night viewing and, or the thermal optic viewing component portions of the embodiment of the device. #51. Included in this drawing of an embodiment thereof. Comprises a switch capable of remotely powering up the reticle portion of the embodiment of the device. #52. Included in this drawing of an embodiment thereof. Comprises a switch capable of remotely

powering up and shutting down the embodiment of the device.

[0125] However, all switching and power distribution module portions of variations of embodiments of the device. As well as all embodiments of the remotely employable controls provided as employable in conjunction with embodiments of the device. Could and may be equipped with any number of additional controls and switching components. Capable of providing the soldier with the ability to remotely activate and control any number of additional functions and service providing abilities, providable by embodiments of the device, remotely employed by same. In this example of an embodiment thereof.

[0126] FIG. 12-B. Includes a drawing of an example of an embodiment, of one of many basic types of common, remotely employable display devices. #64. Equipped with wireless digital video image signal receivers and receiving antennas. As well as of one of many basic types of wireless remotely employable sets of controls, employable in conjunction with embodiments of the device, provided with wireless digital video image transmitting and wireless digital receiving components. With this particular embodiment thereof comprising a digital display device and the set of controls, that are combined as a single portable unit. Employable in conjunction with remotely controllable military equipment and devices. Of which would also include other types and variations of wireless remotely employable sets of controls, equipped with a wireless digital signal transmitter and transmitting antenna, as well as other types and variations of wireless remotely employable display devices. Equipped with wireless digital video image signal receivers and receiving antennas. Including both types and variations of them that are included as an integral portions of each other. As well as other types and variations of them that are separate from, or even portions of other equipment employed by our armed forces. Employable in conjunction with embodiments of the device. Included in this application.

[0127] However regardless of the actual type or variation, of remotely employable sets of controls and display devices, employed by our armed forces, will provide the soldier or squad employing the embodiments of them, with all of the services and abilities, providable by the embodiment of the device, remotely controlled by same, from positions of cover and totally out of the possible line of incoming fire, while doing so, in a broad range of battlefield scenarios and theaters of combat. Previously responsible for a large percentage of U.S. Casualties.

[0128] This drawing includes an embodiment of a very common, type of wireless, remotely employable display device #64. Comprising both an internal wireless receiver. (Not viewable in this drawing of an embodiment thereof.) As well as an omnidirectional wireless receiving antenna. Represented as. #62. In this drawing of an embodiment thereof. Capable of receiving digital video images captured by the digital video image sensor portion of an embodiment of the device and wirelessly transmitted to same, by means of the wireless transmitter and transmitting antenna portion of the device. remotely employed by same.

[0129] This embodiment of a combination. Wireless digital display device and wireless, set of controls. Provided for an embodiment of the device, also includes an omnidirectional wireless digital transmitting antenna. Represented as #63. Included In this drawing of an embodiment thereof. Capable of wireless transmitting digital signals, relative to choices made in the employment of the switching and control component portions of the controls. Included in this drawing of an embodiment thereof, to a wireless receiver and power distribution component portions of the embodiment of the device. Remotely employed by same. Capable of receiving wireless digital signals from the remotely employed controls for the embodiment of the device and providing electrical communication between a power supply for the device and all function enabling component portions of the device, needful of same. Relative to the digital signals received from the transmitting antenna #63. Portion of the controls for the embodiment of the device, remotely employed by the soldier, employing same. Of which is included as integral portions of the embodiment of a display device. Included in this drawing of an embodiment thereof.

[0130] The basic composition of controls. Included in this drawing of an embodiment, of a

remotely employable wireless display device and set of controls. Employable in conjunction with an embodiment of the device. Included in this application. Like the embodiment thereof. Included in FIG. 13. Includes only a small portion of the actual function enabling remotely employable controls, additionally includable in embodiments of the sets of controls for embodiments of the device. Included in this application. However, the set of actual controls. Included in this drawing of an embodiment thereof. For easy reference's sake. Both includes controls of the same types and configurations and capable of controlling the same tasks performable by an embodiment of the device, as is include in the set of controls, included in FIG. 13. Such as switches #51, #52, #53 and switch #54. This also holding true for control knob #55. Also included in this drawing of an embodiment thereof. Capable of providing the soldier remotely controlling an embodiment of the device with the ability to remotely control all vertical and horizontal angular positioning of the embodiment of the device, by means of the pivotable inner portion #57. of this embodiment of a control knob #55. Capable of remotely adjusting the vertical angle of the device, upward, by means of variable compressive pressure applied to the upper portion of #57. And the remotely adjustability of the vertical angle of the device, downward, by means of variable compressive pressure applied to a lower portion of #57. This also being the same for the embodiment of a rotatable outer ring portion #61. Of #55. Comprising an embodiment of a 360-degree horizontal angular adjustment portion of the controls, for the embodiment of the device. Remotely controlled wirelessly, by same. Are represented as controllable, by means of the clockwise and, or counterclockwise rotation of the outer ring. #61. Portion of this embodiment of #55. Also included in this drawing. Additional switching and control portions of this embodiment of a remotely employable, combination of a wireless display device and set of controls. Employable in conjunction with a wirelessly employable embodiment of the device. Included in this application. That are included in this drawing of an embodiment thereof. Yet are not included in the drawing of the set of controls. Included in FIG. 13. For an embodiment of the device. Simply includes a couple, of additional controls, added to the set of controls. Included in this drawing of an embodiment thereof. Such as #51-A. Simply comprising an embodiment of a rotatably adjustable rheostat, control knob. Capable of controlling recital luminosity. Also additionally included in this drawing of an embodiment thereof. Includes an embodiment of a rotatably adjustable, variable control knob. #58. Capable of remotely adjusting the angle of view and level of magnification providable by an embodiment of a digital video image sensor, equipped with electrically controllable, variable optical zooming abilities. Included in an embodiment, of the device. Remotely operated and controlled by this embodiment of a wireless set of controls and digital display device, for an embodiment of the device. In this example of an embodiment thereof.

[0131] The descriptions set forth above are meant to be illustrative and not limiting. Various modifications to the disclosed embodiments, in addition to those described herein, will be apparent to those skilled in the art from the foregoing description. Such modifications are also intended to fall within the scope of the concepts described herein. The disclosures of each patent, patent application, and publication cited or described in this document are hereby incorporated herein by reference, in their entireties. The foregoing description of possible implementations consistent with the present disclosure does not represent a comprehensive list of all such implementations or all variations of the implementations described. The description of some implementations should not be construed as an intent to exclude other implementations described. For example, artisans will understand how to implement the disclosed embodiments in many other ways, using equivalents and alternatives that do not depart from the scope of the disclosure. Moreover, unless indicated to the contrary in the preceding description, no particular component described in the implementations is essential to the invention. It is thus intended that the embodiments disclosed in the specification be considered illustrative, with a true scope and spirit of invention being indicated by the claims when provided

Claims

1. A device for mounting a firearm on a vehicle, the device comprising: a mounting base comprising a base portion capable of being attached to the vehicle, and a rotating portion capable of rotating relative to the base portion; a housing portion disposed on the rotating portion of the mounting base; a first and second bracket capable of being secured to the firearm on opposite lateral sides of the firearm; wherein the first and second bracket are secured to the housing portion.
2. The device of claim 1 wherein the mounting base further comprises a crown gear capable of being turned by a shaft to effectuate the rotation of the rotating portion.
3. The device of claim 1 wherein at least one of the first and second bracket comprise a second rotating portion capable of rotating the firearm the relative to housing portion.
4. The device of claim 1 further comprising a camera affixed to the firearm, wherein the camera is aligned to capture images in a field of view centered in the firearm's line of fire.
5. The device of claim 4, wherein field of view provided by the camera affixed to the firearm is capable of also providing operating visuals for the vehicle.
6. The device of claim 1 further comprising a transmitter capable of transmitting images captured by the camera to a remote display.
7. The device of claim 1 further comprising a set of controls comprising a transmitter capable of transmitting signals to the device providing a user with remote control of both the device and the vehicle.
8. An angularly positionable sighting and surveilling device for a firearm mountable on military equipment comprising: an angularly positionable firearm mount capable of being attached to a military equipment; a camera comprising a field of view, the camera attached to and in alignment with a barrel of the firearm, such that the camera is capable of providing sighting visuals for the firearm therefrom; and a transmitter capable of transmitting images captured by the camera to a remote display.
9. The angularly positionable sighting and surveilling device of claim 8, wherein field of view provided by the camera attached to and in alignment with the barrel of the firearm is additionally capable of providing operating visuals for the military equipment in which the device is employed.
10. The angularly positionable sighting and surveilling device of claim 9, wherein the angularly positionable sighting and surveilling device for a firearm additionally comprises with a set of controls capable of providing a user with the remote control over the angularly positionable sighting and surveilling device and the military equipment.
11. The angularly positionable sighting and surveilling device of claim 8, additionally comprising a mounting base portion.
12. The angularly positionable sighting and surveilling device of claim 11, additionally comprising a first rotatable portion attached to the mounting base portion of the device and capable of providing horizontal angular adjustability to the angularly positionable firearm mount relative to the mounting base portion of the device
13. The angularly positionable sighting and surveilling device of claim 12, additionally comprising a second rotatable portion attached to the first rotatable portion.
14. The device of claim 13, wherein the second rotatable portion is capable of providing vertical angular adjustability to the angularly positionable firearm device relative to the mounting base portion of the device and the military equipment in which its employed.
15. A targeting and surveilling system for remotely controllable military equipment comprising: a remotely controllable firearm mount capable of being mounted on a military equipment, and further capable of providing angular adjustability of a firearm mounted within the remotely controllable firearm mount; a camera comprising a field of view, the camera further being attached to the firearm and in alignment with the barrel of the firearm, such that the camera is capable of providing

both the sighting visuals for the firearm and the operating visuals for the military equipment; and a set of controls capable of providing a user with the combined remote controllability both of the firearm mount and the military equipment on which the mount is mounted.

16. The targeting and surveilling system of claim 15, additionally comprising a transmitter capable of receiving and transmitting video images provided by the camera.

17. The targeting and surveilling system of claim 15, additionally comprising a display capable of receiving and displaying video images provided by the transmitter.

18. The targeting and surveilling system of claim 15, wherein the remotely controllable firearm mount additionally comprises a first rotatable portion capable of providing horizontal angular adjustability to the angularly positionable firearm mount.

19. The system of claim 18, wherein the firearm and the camera are disposed upon a second rotatable portion of the mount, capable of providing vertical angular adjustability to the firearm and camera.
